LESSON 3.6

Ratio Reasoning: Convert Measurements within the Same System

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert between units within the same measurement system.

WHY THIS MATTERS Chef Reyes writes every recipe in the units the supplier uses, but the kitchen's measuring tools are marked in different units of the same system. A stock order lists quarts; the pot is marked in cups. A height rule at the county fair ride is posted in inches; the sign-up sheet lists feet. Your job is to convert between units without changing the amount. The stock order lists 3 quarts, but the pot is marked in cups. The height rule at the ride says 48 inches, but the sign-up sheet lists feet.

REMEMBER

Ratio Reasoning: Convert Measurements within the Same System

Converting Measurements (Same System). Formula: Quantity × (Unit Conversion Ratio) = New Quantity Two measurements can only be compared when they are written in the same unit. Converting uses a special ratio — a unit ratio — that says how much of the smaller unit fits into exactly 1 of the larger unit.

1. Identify the conversion factor (e.g. 1 ft = 12 in, 1 km = 1,000 m).
2. Larger to smaller unit: Multiply by the conversion factor.
3. Smaller to larger unit: Divide by the conversion factor

Words: Conversion factor *(Factor de conversión)* · Unit ratio *(Razón unitaria)* · Convert *(Convertir)* · Measurement system *(Sistema de medidas)* · Double number line *(Recta numérica doble)*

1 **Circle the letter of the best answer. Show how you know.**

Chef Reyes's spice bin holds 48 teaspoons of salt. Her recipe card measures salt in tablespoons, and 3 teaspoons = 1 tablespoon. How many tablespoons of salt are in the bin?

(A) 144 tablespoons
(B) 16 tablespoons
(C) 45 tablespoons
(D) 51 tablespoons

HINT Write the ratio 3 teaspoons : 1 tablespoon.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Chef Reyes needs 2.5 gallons of soup for the fair. There are 4 quarts in 1 gallon and 2 pints in 1 quart. How many pints of soup is that?

(A) 10 pints
(B) 5 pints
(C) 20 pints
(D) 8.5 pints

HINT First change gallons to quarts.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Recipe A uses 2 lemons for 4 cups of water. Recipe B uses 3 lemons for 4 cups of water. Which recipe is more lemony?

(A) Recipe A
(B) They are the same
(C) Recipe B
(D) You cannot tell

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which ratio is greater: 3:4 or 5:8?

(A) 5:8
(B) 3:4
(C) They are equal
(D) Cannot determine

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Chef A uses 2 cups of cheese for every 8 crackers. Chef B uses 3 cups of cheese for every 9 crackers. Who uses more cheese per cracker?

(A) Chef A
(B) Chef B
(C) They use the same amount
(D) Cannot determine

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Which lemonade recipe is more lemony? Recipe A: 2 lemons for 6 cups of water. Recipe B: 3 lemons for 7 cups of water.

(A) Recipe B
(B) Recipe A
(C) They are the same
(D) Cannot determine

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does converting to a smaller unit always make the NUMBER bigger, even though the amount stays exactly the same?

SENTENCE STARTER When the unit gets ___, it takes ___ of them to make the same amount, so I ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.6

Ratio Reasoning: Convert Measurements within the Same System

Set B · Version B · Second form · core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use ratio reasoning to convert between units within the same measurement system.

WHY THIS MATTERS Chef Reyes writes every recipe in the units the supplier uses, but the kitchen's measuring tools are marked in different units of the same system. A stock order lists quarts; the pot is marked in cups. A height rule at the county fair ride is posted in inches; the sign-up sheet lists feet. Your job is to convert between units without changing the amount. The stock order lists 3 quarts, but the pot is marked in cups. The height rule at the ride says 48 inches, but the sign-up sheet lists feet.

1 **Circle the letter of the best answer. Show how you know.**

Chef Reyes is making stock. The recipe calls for 3 quarts of broth, but the only measuring tool on the line is a 1-cup scoop. There are 4 cups in 1 quart. How many inches are in 4 feet?

(A) 3 inches
(B) 16 inches
(C) 44 inches
(D) 48 inches

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Chef Reyes is making stock. The recipe calls for 3 quarts of broth, but the only measuring tool on the line is a 1-cup scoop. There are 4 cups in 1 quart. A bottle holds 500 milliliters. How many bottles make 4 liters?

(A) 4
(B) 8
(C) 500
(D) 2,000

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Chef Reyes is making stock. The recipe calls for 3 quarts of broth, but the only measuring tool on the line is a 1-cup scoop. There are 4 cups in 1 quart. You convert 96 ounces to pounds (16 ounces = 1 pound). Should the number get bigger or smaller?

(A) Bigger, because 96 is already large
(B) Smaller, because pounds are the larger unit
(C) It stays the same
(D) You cannot tell without the ratio table

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How many inches are in 2 feet?

(A) 10
(B) 12
(C) 14
(D) 24

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

How many liters is 2,000 milliliters?

(A) 2 L
(B) 20 L
(C) 200 L
(D) 2,000 L

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Which pair belongs to the SAME measurement system?

(A) Cups and quarts
(B) Inches and centimeters
(C) Pounds and grams
(D) Liters and gallons

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Why does converting to a smaller unit always make the NUMBER bigger, even though the amount stays exactly the same?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help